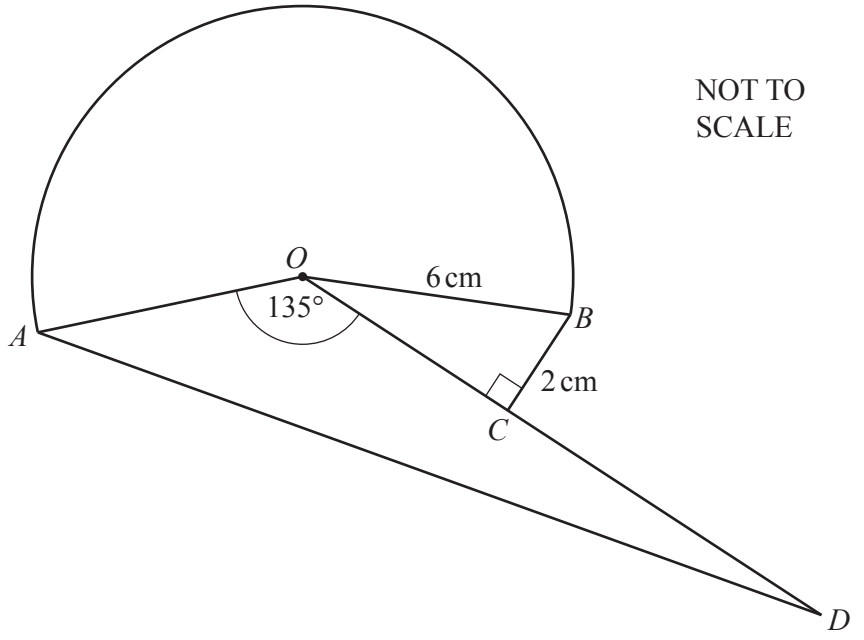
**0580_w23_qp_43_Q8**

**8 (a)**




The diagram shows a shape made from a major sector $AOB$ and triangles $OBC$ and $AOD$.
$OB = 6\,\text{cm}$, $BC = 2\,\text{cm}$, obtuse angle $AOC = 135°$ and angle $BCO = 90°$.

**(i)** Show that angle $BOC = 19.5°$, correct to 1 decimal place.

[2]

**(ii)** Calculate the area of the major sector $AOB$.

..... $\text{cm}^2$ [3]

**0580_w23_qp_43_Q8**

**(iii)** $C$ is the midpoint of $OD$.

Calculate $AD$.

..... cm [5]

**(iv)** Calculate the total area of the shape.

..... $cm^2$ [4]

**(b)** A sector of a circle has radius 8 cm and area 160 $cm^2$.
A mathematically similar sector has radius 20 cm.

Calculate the area of the larger sector.

..... $cm^2$ [3]